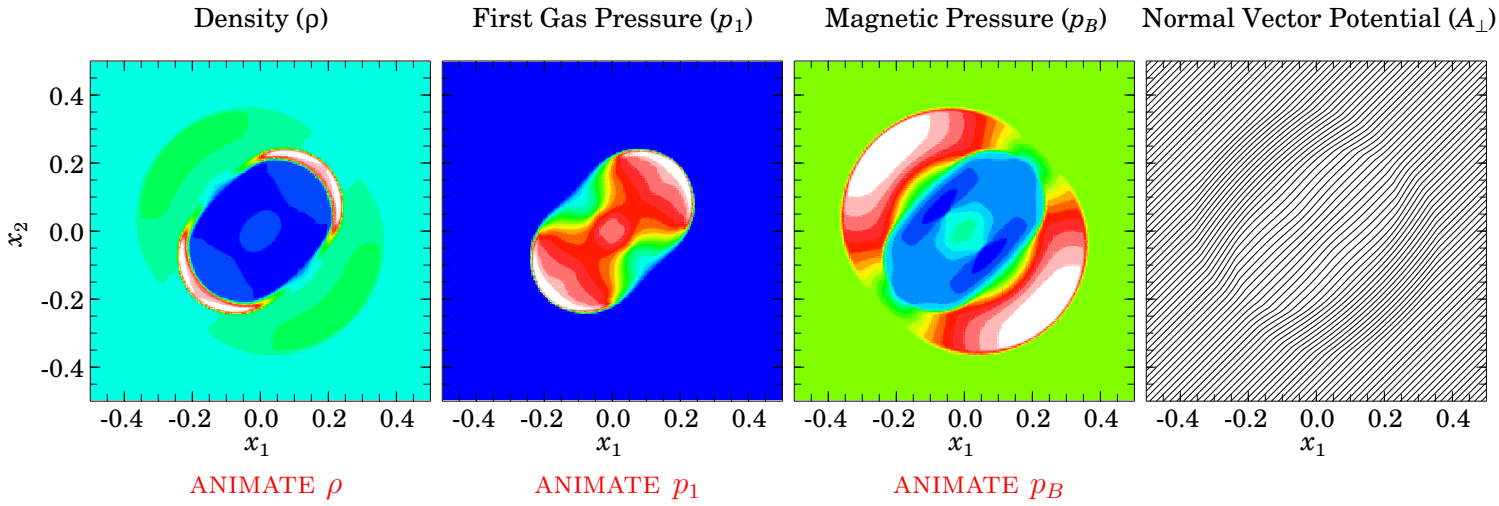
# *ZEUS-3D* 2-D Gallery #1: MHD blast




This problem corresponds to Fig. 13 from Gardiner & Stone (2005, J. Comput. Phys., 205, 509), showing the solution of the 2-D Cartesian MHD blast wave at time $t = 0.02$. The initial state consists of a central disc ($r = 0.125$ centred on the origin) with $(\rho, B_1, B_2, p_1) = (1, 5\sqrt{2}, 5\sqrt{2}, 100)$ embedded in a background state $(1, 5\sqrt{2}, 5\sqrt{2}, 1)$ with $\vec{v} = 0$, $B_3 = 0$ and $\gamma = 5/3$. In the colour panels, white indicates high values, blue low. Contours in the vector potential (far right) are magnetic field lines. The hyperlinks in red open up animations with slightly different colour palettes (*e.g.*, high is red, not white).

The `dzeus35` solution uses 200 by 200 zones, `CMoC`, the total energy equation, and third-order interpolation with the contact steepener engaged. `dzeus35` parameters controlling the time step and artificial viscosity are: `courno=0.5`, `qcon=1.0`, and `qlin=0.1`.

Gardiner & Stone's (2005) 2-D upwinded (Godunov) and *unsplit* scheme (*ATHENA*) uses a staggered mesh for the magnetic field components in which the magnetic compressional derivatives ($\partial_i B_i$) are accounted for. Most multi-dimensional Godunov schemes are constructed from orthogonal sweeps of 1-D seven-wave upwinded schemes which, by design, ignore the compressional terms since they are zero—in 1-D—by the solenoidal condition. As Gardiner & Stone demonstrate, ignoring the compressional terms in 2-D can cause severe grid dependencies in certain 2-D simulations such as this MHD blast problem.

As the *ZEUS* family of codes have *always* included the compressional magnetic terms, the grid-dependent effects discussed by Gardiner & Stone (2005) are absent in the MHD blast wave depicted in the `dzeus35`-generated images and animations above. The end-states look identical to those published by Gardiner & Stone though, quantitatively, there are some differences as seen in the table below where the extrema of the variables are listed. Extrema for the internal energy equation solution ($e_1$) are also given, with the greatest departure from the total energy equation solution ($e_T$) being the density maximum.

Finally, I note that both the total energy and internal energy `dzeus35` solutions (latter not shown) show each of the leading density fronts to be bifurcated into two peaks with a local minimum between them. This is true regardless of angle of inclination of the background magnetic field. It is impossible to tell from their published plot whether the density fronts in Gardiner & Stone are similarly bifurcated.

*Extrema of $\rho$, $p_1$, and $p_B$ in the MHD blast after $t = 0.02$.*

| | $\rho_{\min}$ | $\rho_{\max}$ | $p_{1\min}$ | $p_{1\max}$ | $p_{B\min}$ | $p_{B\max}$ |
|---|---|---|---|---|---|---|
| `dzeus35`, $e_1$ | 0.184 | 4.35 | 0.720 | 31.1 | 24.7 | 75.9 |
| `dzeus35`, $e_T$ | 0.201 | 3.21 | 0.764 | 31.7 | 24.8 | 75.8 |
| *ATHENA* | 0.192 | 3.31 | 1.00 | 32.1 | 23.5 | 77.7 |